Lecture 30

Uniform convergence and power series

Uses.

Summary 2: Every nonempty real set bounded above has a least upper bound.
Summary 3: Sequence limits and geometric limits obey arithmetic laws.
Summary 4: Bounded monotone sequences converge, squeeze holds, and $n^{1/n} \to 1$.
Summary 5: Continuity is defined relative to a function's domain; elementary continuous functions remain continuous under finite sums.
Summary 8: The elementary derivatives and chain rule hold.
Summary 25: Finite integrals are linear and order preserving, with $|\int f| \le \int |f|$.
Summary 26: Continuous functions on closed finite intervals are Riemann integrable. The fundamental theorem differentiates their accumulation integrals.
Summary 29: Partial sums define convergence. Absolute convergence, geometric comparison, ratio/root tests and the alternating remainder bound hold.

Objectives. *For functions, convergence at each point can leave a large error elsewhere in the domain. Uniform convergence controls that error with one index. It preserves continuity and permits finite-interval integration of a series. A power series has this control on each closed interval strictly inside its radius.*

By the end of this lecture you will be able to:

- Distinguish pointwise and uniform convergence by their quantifiers and a counterexample.
- Prove the majorant test and continuity of a uniform limit.
- Justify termwise integration of continuous series on a closed finite interval.
- Determine power-series radii and endpoints, with uniform convergence on interior closed intervals.

30.1 One index for the whole domain

Let $E \subseteq \mathbb{R}$ be nonempty and let $f_n : E \to \mathbb{R}$ be functions. They converge *pointwise* to $f : E \to \mathbb{R}$ if for each $x \in E$, the numerical sequence $f_n(x)$ tends to $f(x)$. Written with quantifiers, for every $x \in E$ and $\eta > 0$, there is an index N, possibly depending on both, such that $|f_n(x) - f(x)| < \eta$ for every $n \ge N$. They converge *uniformly* on E if

$$\text{for every } \eta > 0 \text{ there is } N \text{ such that } |f_n(x) - f(x)| < \eta \quad \text{for all } n \ge N \text{ and all } x \in E.$$

The index is independent of x. Uniform convergence implies pointwise convergence by using that same index at any one point. A series of functions converges pointwise or uniformly when its sequence of finite partial sums does so.

Example 30.1 (A moving large error). Consider $f_n(x) = x^n$ on $E = [0,1]$. For each fixed $x < 1$, the geometric limit gives $f_n(x) \to 0$; at $x = 1$, the value is always 1. The pointwise limit is therefore

$$f(x) = \begin{cases} 0, & 0 \le x < 1, \\ 1, & x = 1. \end{cases}$$

For every $n \ge 1$, choose $x_n = (1/2)^{1/n} \in (0,1)$. Then $f(x_n) = 0$ but $f_n(x_n) = 1/2$. For tolerance $1/4$, every proposed uniform index fails at some such point. Thus convergence is not uniform on $[0,1]$. On $[0,r]$ for any fixed $0 < r < 1$, however, $|f_n(x)| \le r^n \to 0$ independently of x, so convergence is uniform. The domain is part of the convergence claim.

Figure 30.1 shows continuous functions whose pointwise limit is discontinuous at 1. The moving witness above proves failure of uniformity.

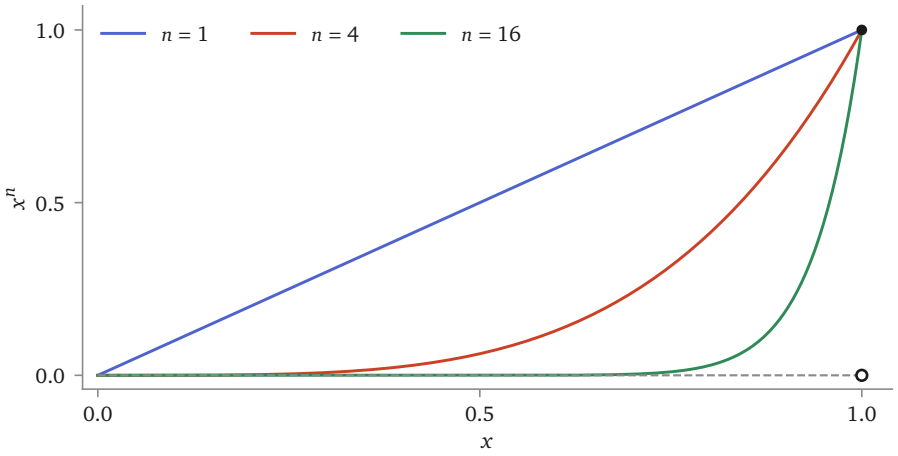


Figure 30.1: The functions x, x^4 and x^{16} on $[0,1]$. Their pointwise limit is zero below 1 and has value 1 at the endpoint.

30.2 A majorant that controls every tail

Theorem 30.2 (Weierstrass majorant test). *Let $f_n : E \to \mathbb{R}$. Suppose numbers $M_n \ge 0$ satisfy $|f_n(x)| \le M_n$ for every $x \in E$ and every $n \ge 1$, and that $\sum_{n=1}^{\infty} M_n$ converges. Then $\sum f_n$ converges absolutely at every point and uniformly on E.*

Proof. Fix $x \in E$. Numerical comparison with $\sum M_n$ gives convergence of $\sum |f_n(x)|$, hence of $\sum f_n(x)$ by Lecture 29. Write its sum as $F(x)$ and its partial sums as $S_N(x)$. For $m > N$, the finite triangle inequality gives

$$|S_m(x) - S_N(x)| \le \sum_{n=N+1}^{m} M_n.$$

Let $m \to \infty$ at this fixed point. Both sides have finite limits, so

$$|F(x) - S_N(x)| \le \sum_{n=N+1}^{\infty} M_n.$$

The numerical majorant tail tends to zero and does not depend on x. One sufficiently large N therefore works for every $x \in E$. This proves uniform convergence, in addition to the already proved absolute convergence. ∎

For example, $\sum_{n=1}^{\infty} \cos(nx)/3^n$ satisfies $|\cos(nx)/3^n| \le 3^{-n}$ on all of $\mathbb{R}$. The geometric majorant converges, so the series converges uniformly and absolutely there. Its uniform error after N terms is at most $\sum_{n=N+1}^{\infty} 3^{-n} = 1/(2 \cdot 3^N)$. The estimate is valid even though the domain is unbounded.

30.3 Continuity and integration

Theorem 30.3 (Continuity of a uniform limit). *If every $f_n : E \to \mathbb{R}$ is continuous and $f_n \to f$ uniformly on E, then f is continuous on E in the relative-domain sense.*

Proof. Fix $a \in E$ and $\eta > 0$. Uniform convergence permits one index n with $|f_n(x) - f(x)| < \eta/3$ for all $x \in E$. Continuity of this one function f_n at a supplies $\delta > 0$ with $|f_n(x) - f_n(a)| < \eta/3$ whenever $x \in E$ and $|x - a| < \delta$. For such points the triangle inequality gives

$$|f(x) - f(a)| \le |f(x) - f_n(x)| + |f_n(x) - f_n(a)| + |f_n(a) - f(a)| < \eta.$$

This is relative continuity at a, and a was arbitrary. ∎

Each finite sum of the preceding cosine series is continuous. The majorant test and this theorem therefore make its sum continuous on $\mathbb{R}$. In contrast, the discontinuous limit in the power example cannot be a uniform limit of those continuous functions on $[0, 1]$.

Theorem 30.4 (Integration of a uniform limit). *Let $a < b$ be finite. If $f_n : [a, b] \to \mathbb{R}$ are continuous and converge uniformly to f, then*

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{n\to\infty} \int_a^b f_n(x)\,\mathrm{d}x.$$

Consequently, a uniformly convergent series of continuous functions may be integrated term by term on $[a, b]$.

Proof. The preceding theorem makes f continuous, so every integral in the statement exists by Lecture 26. Given $\eta > 0$, uniform convergence eventually gives $|f_n - f| < \eta/(b - a)$ everywhere on $[a, b]$. Linearity, the absolute-value bound and order imply

$$\left| \int_a^b f_n - \int_a^b f \right| \le \int_a^b |f_n - f| \le \eta.$$

To obtain a strict error bound for any prescribed tolerance, choose half that tolerance as η. This proves the limit. For a series, apply the result to its continuous partial sums and use finite linearity. This proves convergence of the numerical series of integrals and identifies its sum with the integral of the function series. ∎

For the uniformly convergent cosine series, termwise integration on $[0, \pi]$ gives

$$\int_0^{\pi} \left(\sum_{n=1}^{\infty} \frac{\cos(nx)}{3^n} \right) \mathrm{d}x = \sum_{n=1}^{\infty} \frac{\sin(n\pi) - \sin 0}{n 3^n} = 0.$$

Every finite summand integral is zero. The uniform-limit theorem justifies passing from those finite calculations to the whole series.

Pointwise convergence alone does not permit the integral exchange. For $n \geq 1$, consider the continuous functions $h_n(x) = n^2 x(1-x)^n$ on $[0,1]$. At both endpoints they are zero. At a fixed interior point, put $q = 1-x$. Then $0 < q < 1$, and the growth limit of Lecture 29 gives

$$h_n(x) = x\big(n(\sqrt{q})^n\big)^2 \longrightarrow 0.$$

Thus the pointwise limit is the continuous zero function on the whole interval. Yet finite substitution $u = 1-x$ and the power primitives give

$$\int_0^1 h_n(x)\,\mathrm{d}x = n^2 \int_0^1 (u^n - u^{n+1})\,\mathrm{d}u = \frac{n^2}{(n+1)(n+2)} \longrightarrow 1.$$

The integral of the pointwise limit is zero. Even continuity of every function and of its limit has not justified exchanging the limit and integral. The missing uniform control is the hypothesis that excludes this example from the theorem.

Uniform convergence alone does not justify differentiation. The functions $g_n(x) = \sin(nx)/n$ converge uniformly to zero on $\mathbb{R}$ because $|g_n(x)| \leq 1/n$. Their derivatives are $\cos(nx)$, with value 1 at zero for every n. They cannot tend there to the derivative 0 of the limit function. Lecture 31 will prove a separate differentiation theorem for power series.

30.4 Power series and Abel's bound

A *power series centred at zero* is a series

$$\sum_{n=0}^{\infty} a_n x^n, \qquad a_n \in \mathbb{R}.$$

Here $x^0 = 1$, also at $x = 0$. A series centred at $c \in \mathbb{R}$ replaces x^n by $(x-c)^n$; every result below applies after setting $y = x - c$. We first work at centre zero.

Theorem 30.5 (Abel's bound). *Suppose $x_0 \neq 0$ and the sequence $(a_n x_0^n)$ is bounded in absolute value by M. If $0 \leq r < |x_0|$, the series converges absolutely and uniformly for $|x| \leq r$.*

Proof. For $|x| \leq r$, the bound at x_0 gives

$$|a_n x^n| = |a_n x_0^n| \left|\frac{x}{x_0}\right|^n \leq M \left(\frac{r}{|x_0|}\right)^n.$$

The right side is a convergent geometric majorant, since $r/|x_0| < 1$. Apply the majorant test, keeping the finite constant term separately if indexing that test from 1. This proves both conclusions. ∎

In particular, convergence at a nonzero point bounds its terms by the term test and implies absolute convergence at every point closer to zero. No conclusion is forced at the same absolute distance.

Define the nonnegative set of bounded-term radii and its extended supremum by

$$B = \{\rho \geq 0 : (a_n \rho^n) \text{ is bounded}\}, \qquad R = \sup B.$$

The set contains zero, because only the constant term survives there. If it is unbounded, set $R = \infty$; otherwise completeness gives its finite supremum. This number is the *radius of convergence*.

Theorem 30.6 (The radius and interior uniform convergence). *A power series converges absolutely for $|x| < R$ and diverges for $|x| > R$. For every $0 \le r < R$, it converges uniformly on $[-r, r]$. When $0 < R < \infty$, each endpoint $x = R$ and $x = -R$ must be tested separately. If $R = 0$, the series still converges at zero; if $R = \infty$, it converges at every real point.*

Proof. For $0 \le r < R$, the supremum definition supplies $\rho \in B$ with $\rho > r$; if $R = \infty$, unboundedness supplies it. Apply Abel's bound at $x_0 = \rho$. This gives absolute and uniform convergence on $[-r, r]$, proving the interior statements. A point with $|x| < R$ lies in such an interval, except when $R = 0$ there are no such points. At zero the series is always the finite value a_0.

If a series converged at a point with $|x| > R$, its terms would tend to zero and hence be bounded. Their absolute values equal $|a_n||x|^n$, so $|x| \in B$, contradicting the supremum. Thus it diverges outside. Abel's strict inequality says nothing about finite nonzero endpoints. The examples below show why separate endpoint tests are necessary. ∎

Each power-series partial sum is a continuous polynomial. On any closed interval inside the radius, uniform convergence therefore makes the sum continuous and permits termwise integration there. Every interior point lies in one such interval, so the sum is continuous throughout $(-R, R)$.

30.5 Integrating a power series

The uniform convergence just proved supplies an explicit primitive, without assuming a termwise differentiation rule.

Theorem 30.7 (Integrated power series). *If $F(x) = \sum_{n=0}^{\infty} a_n x^n$ has radius R, then for $|x| < R$,*

$$G(x) := \int_0^x F(t)\,\mathrm{d}t = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}, \qquad G'(x) = F(x).$$

The series on the right has the same radius R.

Proof. Fix $|x| < r < R$. The original series converges uniformly on $[-r, r]$, and its sum is continuous there. Apply the uniform integration theorem to its finite partial sums between 0 and x, using endpoint reversal if $x < 0$. The power primitive gives the displayed coefficients. At $x = 0$, the integral is zero by the equal-endpoint convention and every series term is zero because its power has positive exponent. The fundamental theorem of Lecture 26 then gives $G' = F$ inside the radius. Every $|x| < R$ is covered by one such interval.

Let R_G be the radius of the integrated series. For $|x| < R$, its absolute terms satisfy $|a_n x^{n+1}|/(n+1) \le |x|\,|a_n x^n|$. Absolute comparison gives $R_G \ge R$. If $R = \infty$, this already identifies both radii. Suppose instead $R < \infty$ and $R_G > R$. Choose real numbers $R < r < \rho < R_G$. Convergence of the integrated series at ρ bounds its terms: $|a_n \rho^{n+1}|/(n+1) \le M$ for some M and every n. It follows that

$$|a_n r^n| \le \frac{M}{\rho}(n+1)\left(\frac{r}{\rho}\right)^n \longrightarrow 0,$$

using both geometric decay and $nq^n \to 0$ from Lecture 29. Thus $(a_n r^n)$ is bounded and $r \in B$, contradicting $r > \sup B = R$. This proves $R_G = R$, including the possibility $R = 0$. ∎

Example 30.8 (A sum from an integrated geometric series). Find $F(x) = \sum_{n=0}^{\infty}(n+1)x^n$ inside its radius. The coefficient ratio $(n+2)/(n+1) \to 1$ and Lecture 29's ratio test give convergence for $|x| < 1$ and divergence for $|x| > 1$, so its radius is 1. The integration theorem constructs

$$G(x) = \sum_{n=0}^{\infty} x^{n+1} = \frac{x}{1-x} \qquad (|x| < 1),$$

where the last equality is the geometric sum. The same theorem gives $F = G'$. The ordinary quotient rule therefore yields $F(x) = 1/(1-x)^2$ on $(-1, 1)$. The argument differentiates an already identified primitive, rather than assuming a derivative rule for arbitrary infinite sums.

30.6 Finding the radius and testing the endpoints

If the coefficients are eventually nonzero and $|a_{n+1}|/|a_n| \to L \in [0, \infty]$, the term ratio at any $x \neq 0$ tends to $|x|L$. The ratio test gives

$$R = \frac{1}{L}, \qquad 1/0 = \infty, \quad 1/\infty = 0.$$

For finite positive L, it proves convergence at every $|x| < 1/L$ and divergence at every $|x| > 1/L$, identifying the radius. For $L = 0$, every finite x is inside; for $L = \infty$, every nonzero x is outside. These give the two stated conventions.

Likewise, if $|a_n|^{1/n} \to K \in [0, \infty]$, with $n \geq 1$, the root test at $x \neq 0$ gives the radius $R = 1/K$ with the same conventions. It allows zero coefficients, since no consecutive-term quotient is needed. The root limits of Lecture 4 must be used when simplifying factors such as $n^{1/n}$. If the required coefficient limit does not exist, the particular formula gives no conclusion; the radius theorem still applies.

Example 30.9 (A shifted series with different endpoints). Find the interval of convergence of

$$\sum_{n=1}^{\infty} \frac{(x-2)^n}{n3^n}.$$

Its coefficient ratio in the variable $y = x - 2$ is $n/(3(n+1)) \to 1/3$, so the radius is 3. The open interval is $-1 < x < 5$. At $x = 5$, the series becomes the divergent harmonic series. At $x = -1$, it becomes $\sum(-1)^n/n$, which converges conditionally by the alternating test. Thus the full interval is $[-1, 5)$. Uniform convergence is guaranteed on every $[2-r, 2+r]$ with $r < 3$; the radius theorem does not itself promise uniform convergence at the endpoints.

A general radius formula is recorded as a cited result [C], the *Cauchy–Hadamard formula*. For a nonnegative sequence u_n, its *limit superior* is the extended limit of the decreasing tail suprema: $\limsup u_n = \lim_{N\to\infty} \sup_{n\geq N} u_n$. Infinite tail suprema are allowed; otherwise their decreasing nonnegative values have a limit by monotone convergence. The cited formula is $R = 1/(\limsup |a_n|^{1/n})$, with the conventions above. The radius proof and all computations here do not require this more general formula.

30.7 Uniform convergence at a conditional endpoint

Consider the series $\sum_{n=1}^{\infty}(-1)^{n-1}x^n/n$ on $[0, 1]$. For each fixed x, the magnitudes $b_n(x) = x^n/n$ are nonincreasing and at most $1/n$, hence tend to zero. The alternating theorem gives a

sum $F(x)$ and the bound

$$|F(x) - S_N(x)| \le \frac{x^{N+1}}{N+1} \le \frac{1}{N+1} \qquad (0 \le x \le 1).$$

This proves uniform convergence even on the closed interval. At $x = 1$, convergence is conditional. No convergent numerical majorant can dominate every absolute summand on this domain, since it would have to dominate $1/n$ at that endpoint. Thus the majorant test is sufficient, and uniform convergence is possible without its hypotheses. The sum is nevertheless continuous on $[0, 1]$ and may be integrated termwise there. Lecture 31 identifies this sum and proves the additional derivative rule.

Summary 30.

- Uniform convergence uses one index for all domain points. Pointwise convergence alone need not preserve continuity, as x^n on $[0, 1]$ shows.
- Theorem 30.2: a convergent numerical majorant bounds every tail uniformly and gives pointwise absolute convergence.
- Theorems 30.3 and 30.4: uniform limits preserve continuity and permit integration of continuous functions on a closed finite interval. Differentiation needs a separate argument.
- Theorems 30.5 and 30.6: the radius gives absolute convergence inside and divergence outside; convergence is uniform on each closed interval strictly inside. Finite nonzero endpoints require separate tests.
- Theorem 30.7: integrating coefficients gives a primitive inside the radius and preserves that radius. Its derivative follows from FTC.
- Coefficient ratio or root limits, when they exist under the stated hypotheses, give $R = 1/L$ for the ratio limit and $R = 1/K$ for the root limit. The general limit-superior formula is cited [C].

Sheet 30 develops further uniform tests, the Dirichlet argument and the arctangent expansion by termwise integration.

References

J. Stewart, D. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th ed., Cengage, 2020, §§11.8–11.9. S. Abbott, *Understanding Analysis*, 2nd ed., Springer, 2015, §§6.2–6.5. For the cited Cauchy–Hadamard formula, see W. Rudin, *Principles of Mathematical Analysis*, 3rd ed., McGraw–Hill, 1976, Theorem 3.39, p. 69.